

L5c: Linear Programming

Allocation, duality, and resource values

CHEME 5800 Cornell University

Objectives for Today

How should we allocate limited resources, and how can we establish that a feasible allocation is optimal?

Today's concepts

- **Formulate allocation models:** identify variables, objectives, constraints, and bounds for purchases and network flows.
- **Explain duality and resource values:** construct a dual bound and interpret the value of additional budget.
- **Evaluate results:** check solver status, feasibility, objective values, and an optimality bound.

Lecture: [Introduction to Linear Programming](#)

Lecture and Example

Lecture: Introduction to Linear Programming

Formulate resource-allocation and network-flow models, then use duality to bound the objective and interpret resource values.

Example: Apples, Oranges, and Linear Allocation

Compare three utility choices with the same prices and budget. Predict the purchases, solve the models, and verify expenditure and utility.

Equal utility per dollar can give many optimal allocations.

Primal Linear Programming Problems

The **primal** describes the activities we choose: purchases, production levels, or edge flows.

For n activities and m resource constraints, define:

- $\mathbf{x} \in \mathbb{R}^n$: activity levels; c_i is the objective contribution per unit of activity i .
- $\mathbf{A} \in \mathbb{R}^{m \times n}$: $A_{j,i}$ is the amount of resource j used per unit of activity i .
- $\mathbf{b} \in \mathbb{R}^m$: b_j is the available amount of resource j .

Matrix rows represent **resources**; columns represent **activities**.

Maximize Value Within Resource Limits

With nonnegative activities, the resource-allocation model is:

$$\begin{aligned} &\underset{\mathbf{x}}{\text{maximize}} && O(\mathbf{x}) = \mathbf{c}^\top \mathbf{x} = \sum_{i=1}^n c_i x_i \\ &\text{subject to} && \mathbf{Ax} \leq \mathbf{b}, \\ &&& \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Each row limits total use of one resource:

$$\underbrace{\sum_{i=1}^n A_{j,i} x_i}_{\text{resource } j \text{ used}} \leq \underbrace{b_j}_{\text{available}}, \quad j = 1, \dots, m.$$

Coefficients are fixed; the contributions from activities add together.

Feasible Does Not Mean Optimal

The **feasible region** contains all decisions satisfying every constraint and bound.

- An **optimal** decision is feasible and attains the best objective value. Several decisions can share that value.
- An **infeasible** model has no decision satisfying all constraints.
- An **unbounded** maximization model has feasible decisions with arbitrarily large objective values, so no finite optimum exists.

Linear programs can also minimize an objective, use equality constraints, and impose other lower or upper bounds.

Consumer Choice as a Linear Program

For product $i = 1, \dots, n$, let $x_i \geq 0$ be the quantity purchased, $p_i > 0$ its price in dollars per unit, and $u_i \geq 0$ its utility per unit. Let $I \geq 0$ be the budget in dollars.

The consumer maximizes total utility subject to the budget:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{maximize}} && U(\mathbf{x}) = \sum_{i=1}^n u_i x_i \\ & \text{subject to} && \sum_{i=1}^n p_i x_i \leq I, \\ & && x_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

One resource, the budget: $\mathbf{c} = \mathbf{u}$, $\mathbf{A} = [p_1 \ \cdots \ p_n]$, and $\mathbf{b} = [I]$.

Assumptions: divisible quantities, constant prices and utility per unit, and sufficient product availability.

Worked Allocation: Apples and Oranges

Let x_A and x_O be the quantities of apples and oranges. Apples cost 2 dollars per unit and provide 0.55 utility units; oranges cost 4 dollars and provide 0.45 utility units per unit.

With a 100-dollar budget, the linear program is:

$$\begin{aligned} & \underset{x_A, x_O}{\text{maximize}} && U(x_A, x_O) = 0.55x_A + 0.45x_O \\ & \text{subject to} && 2x_A + 4x_O \leq 100, \\ & && x_A, x_O \geq 0. \end{aligned}$$

We can afford at most 50 apples or 25 oranges. Affordable mixtures are also feasible.

Compare Purchases Along the Budget Boundary

Both fruits provide positive utility. With divisible quantities and no stock limit, spending unused budget improves utility. Thus, an optimum uses the full budget:

$$x_O = 25 - \frac{1}{2}x_A, \quad 0 \leq x_A \leq 50.$$

Substituting into utility gives:

$$\begin{aligned} U(x_A) &= 0.55x_A + 0.45 \left(25 - \frac{1}{2}x_A \right) \\ &= 11.25 + 0.325x_A. \end{aligned}$$

Utility increases with x_A . The optimum is $(x_A^*, x_O^*) = (50, 0)$, with $U^* = 27.5$. Buying only oranges gives 11.25 utility units.

Compare Utility per Dollar

The ratio u_i/p_i measures utility gained per dollar spent on product i . For the first case:

$$\frac{u_A}{p_A} = \frac{0.55}{2} = 0.275, \quad \frac{u_O}{p_O} = \frac{0.45}{4} = 0.1125.$$

- Moving one dollar from oranges to apples increases utility.
- Spending the whole budget does not, by itself, establish optimality.
- Changing prices or utilities can change the preferred fruit.

This comparison will also explain the value of an additional dollar.

Comparing Allocations and Their Geometry

Keep prices $(p_A, p_O) = (2, 4)$ and budget $I = 100$ fixed. Changing utility coefficients gives three cases:

Case	(u_A, u_O)	$(u_A/p_A, u_O/p_O)$	Optimal purchase	Utility
A	(0.55, 0.45)	(0.275, 0.1125)	(50, 0)	27.5
B	(0.15, 0.55)	(0.075, 0.1375)	(0, 25)	13.75
C	(2, 4)	(1, 1)	Any full-budget mix	100

The larger ratio receives the budget. Equal ratios make every dollar equally valuable, whichever fruit we purchase.

In Case C, (50, 0), (0, 25), and (20, 15) all give 100 utility units.

Compare the Budget and Utility Slopes

Put apples on the horizontal axis and oranges on the vertical axis. Let $\bar{U}$ be a fixed total utility, m_I the budget slope, and m_O the objective-line slope.

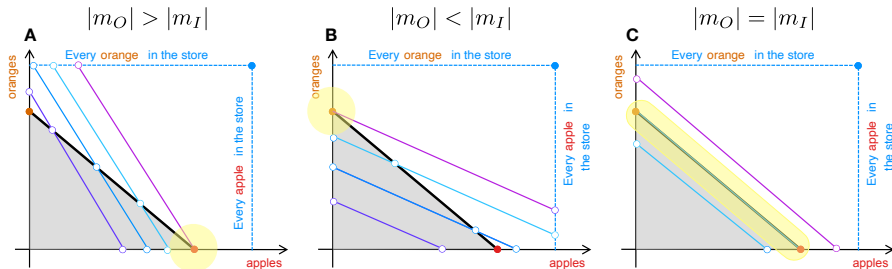
For positive prices and utilities, the lines are:

$$\text{Budget: } x_O = \frac{I}{p_O} + \underbrace{\left(-\frac{p_A}{p_O}\right)}_{m_I} x_A,$$

$$\text{Utility: } x_O = \frac{\bar{U}}{u_O} + \underbrace{\left(-\frac{u_A}{u_O}\right)}_{m_O} x_A.$$

Increasing $\bar{U}$ shifts the utility line upward. Seek the last line that still touches the feasible region.

The Optimum Can Be a Corner or an Edge



- **A:** $|m_O| > |m_I|$ gives the apple corner.
- **B:** $|m_O| < |m_I|$ gives the orange corner.
- **C:** $|m_O| = |m_I|$ gives an entire optimal budget edge.

Schematic: the gray triangle is feasible; dashed stock limits lie beyond affordable quantities.

Minimum-Cost Network Flow

Several routings can deliver the same amount of flow at different costs. We now seek the **least expensive routing of a required delivery**.

Use a finite directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, distinct source s and sink t , and no self-loops. Fix vertex and edge order.

- f_j : flow on edge j ; $c_j \geq 0$: its finite capacity.
- w_j : cost per unit of flow on edge j .
- $F \geq 0$: the specified net flow from source to sink.

Flows, capacities, and F use the same units.

An Incidence Matrix Encodes Conservation

The matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{E}|}$ has one row per vertex i and one column per edge j . Its entries and required net inflows b_i are:

$$A_{i,j} = \begin{cases} 1, & j \text{ enters } i, \\ -1, & j \text{ leaves } i, \\ 0, & \text{otherwise.} \end{cases} \quad b_i = \begin{cases} -F, & \text{source,} \\ F, & \text{sink,} \\ 0, & \text{otherwise.} \end{cases}$$

Collect flows in $\mathbf{f} \in \mathbb{R}^{|\mathcal{E}|}$ and net inflows in $\mathbf{b} \in \mathbb{R}^{|\mathcal{V}|}$. Each row enforces:

$$\underbrace{(\mathbf{A}\mathbf{f})_i}_{\text{inflow minus outflow}} = b_i.$$

A Linear Program for the Required Delivery

Encode the required flow F in $\mathbf{b}$ and minimize total edge cost:

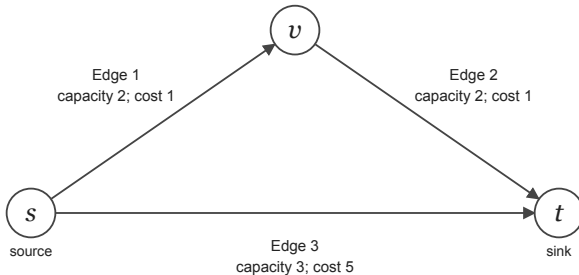
$$\begin{aligned} & \underset{\mathbf{f}}{\text{minimize}} && \sum_{j \in \mathcal{E}} w_j f_j \\ & \text{subject to} && \mathbf{A}\mathbf{f} = \mathbf{b}, \\ & && 0 \leq f_j \leq c_j, \quad j \in \mathcal{E}. \end{aligned}$$

- Equalities enforce delivery and conservation.
- Bounds enforce nonnegative flows and edge capacities.
- Costs distinguish feasible routings with the same delivery.

This formulation applies to general directed networks.

Worked Network: Two Routes

Deliver $F = 3$ flow units. Costs are in dollars per flow unit.



Through v : cost $1 + 1 = 2$ dollars per unit, capacity 2. Direct: cost 5 dollars per unit.

Some of the 3-unit delivery must travel directly.

Write the Three Vertex Balances

Order rows as (s, v, t) and columns as edges $(1, 2, 3)$. For $F = 3$, conservation gives:

$$\underbrace{\begin{bmatrix} -1 & 0 & -1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}}_{\mathbf{f}} = \underbrace{\begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}.$$

- Source: $f_1 + f_3 = 3$ units leave.
- Intermediate vertex: $f_1 = f_2$.
- Sink: $f_2 + f_3 = 3$ units enter.

Each edge column contains a -1 at its start and a $+1$ at its end.

Use the Cheaper Route Up to Its Capacity

Let q be the flow through v . Conservation gives $f_1 = f_2 = q$ and $f_3 = 3 - q$; capacities require $0 \leq q \leq 2$.

The total cost, in dollars, is:

$$C(q) = \underbrace{q + q}_{\text{route through } v} + \underbrace{5(3 - q)}_{\text{direct route}} = 15 - 3q.$$

Cost decreases with q , so use $q^* = 2$:

$$(f_1^*, f_2^*, f_3^*) = (2, 2, 1), \quad C^* = 9.$$

Sending all 3 units directly is feasible, but costs 15 dollars. Conservation and capacity checks alone do not select the cheapest routing.

Minimum-Cost Maximum Flow Takes Two Steps

Let $F_{\max}$ be the largest feasible net source-to-sink flow. To find a minimum-cost maximum flow:

1. Solve the maximum-flow problem to find $F_{\max}$.
2. Set $F = F_{\max}$ and minimize cost among routings delivering it.

In our network, source capacities give a bound reached by a feasible flow:

$$F_{\max} = 2 + 3 = 5, \quad (f_1, f_2, f_3) = (2, 2, 3).$$

Delivering 5 units costs 19 dollars; the earlier 9-dollar result delivers 3 units. A requirement above 5 units is infeasible.

Next lab: [L5d uses costs and capacities in a worker-task assignment model.](#)

Dual Linear Programming Problems

How can we show that no feasible purchase is better? Assign a value $y \geq 0$ to each dollar of budget, in utility units per dollar.

Recall product prices p_i , utilities u_i , and budget I . The **consumer dual** finds the smallest utility bound:

$$\begin{aligned} & \underset{y}{\text{minimize}} && Iy \\ & \text{subject to} && p_i y \geq u_i, \quad i = 1, \dots, n, \\ & && y \geq 0. \end{aligned}$$

Each constraint makes the assigned value of a product's cost cover its utility. A feasible purchase spends at most I , so its utility is at most Iy .

What Is the Consumer's Budget Worth?

For our apple-orange case, $2y \geq 0.55$ and $4y \geq 0.45$. The smallest feasible value and its budget bound are:

$$y^* = \max\left\{\frac{0.55}{2}, \frac{0.45}{4}\right\} = 0.275, \quad Iy^* = 27.5.$$

Buying 50 apples attains 27.5 utility units. Matching the bound **proves optimality**.

The budget's **shadow price** is y^* : an extra dollar buys 0.5 units of apples and adds 0.275 utility units.

Interpretation: fixed prices and utilities, divisible quantities, and sufficient stock. Shadow prices can change as resource availability changes.

The General Primal-Dual Pair

Return to n activities and m resources, with $\mathbf{A} \in \mathbb{R}^{m \times n}$. Let $\mathbf{y} \in \mathbb{R}^m$ assign a value to each resource. The paired models are:

Primal: choose activities

$$\begin{aligned} \max_{\mathbf{x}} \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} \leq \mathbf{b}, \\ & \mathbf{x} \geq \mathbf{0}. \end{aligned}$$

Dual: value resources

$$\begin{aligned} \min_{\mathbf{y}} \quad & \mathbf{b}^\top \mathbf{y} \\ \text{s.t.} \quad & \mathbf{A}^\top \mathbf{y} \geq \mathbf{c}, \\ & \mathbf{y} \geq \mathbf{0}. \end{aligned}$$

There is one dual variable per primal resource constraint and one dual constraint per primal activity.

Resource Values Must Cover Each Activity

Recall that $A_{j,i}$ is resource j used per unit of activity i , and c_i is that activity's objective contribution. Value resource j at $y_j \geq 0$ objective units per resource unit.

For each activity, the dual requires:

$$\underbrace{\sum_{j=1}^m A_{j,i} y_j}_{\text{assigned resource value}} \geq \underbrace{c_i}_{\text{contribution}}, \quad i = 1, \dots, n.$$

The transpose combines entries down each activity column of $\mathbf{A}$.

The objective $\mathbf{b}^\top \mathbf{y}$ totals the assigned value of all available resources. The dual seeks the smallest such bound.

Weak Duality Bounds Every Feasible Decision

Let $\mathbf{x}$ be primal-feasible and $\mathbf{y}$ dual-feasible. Their constraints and nonnegative components give:

$$\mathbf{c}^\top \mathbf{x} \leq (\mathbf{A}^\top \mathbf{y})^\top \mathbf{x} = \mathbf{y}^\top \mathbf{A} \mathbf{x} \leq \mathbf{y}^\top \mathbf{b}.$$

- First inequality: $\mathbf{A}^\top \mathbf{y} \geq \mathbf{c}$ and $\mathbf{x} \geq \mathbf{0}$.
- Last inequality: $\mathbf{A} \mathbf{x} \leq \mathbf{b}$ and $\mathbf{y} \geq \mathbf{0}$.

This is **weak duality**: every feasible dual objective is an upper bound on every feasible primal objective.

A Zero Duality Gap Certifies Optimality

For feasible candidates, define the **duality gap** as:

$$\mathbf{b}^\top \mathbf{y} - \mathbf{c}^\top \mathbf{x} \geq 0.$$

A zero gap proves both candidates are optimal: the primal reaches the upper bound supplied by the dual.

Strong duality: if the primal is feasible with a finite optimal value, the dual also has an optimal solution. Optimal solutions satisfy:

$$\mathbf{c}^\top \mathbf{x}^\star = \mathbf{b}^\top \mathbf{y}^\star.$$

The objective values agree; the decision vectors describe different quantities.

Constraint Directions and Variable Signs

For the **maximization convention** just used, the bound determines the following correspondences:

Primal condition

Corresponding dual condition

$$\sum_i A_{j,i} x_i \leq b_j$$

$$y_j \geq 0$$

$$\sum_i A_{j,i} x_i = b_j$$

y_j free to have either sign

$$x_i \geq 0$$

$$(\mathbf{A}^\top \mathbf{y})_i \geq c_i$$

x_i free to have either sign

$$(\mathbf{A}^\top \mathbf{y})_i = c_i$$

An equality can be multiplied by either sign. A free primal variable requires matching coefficients for both positive and negative values.

Network-flow equality balances therefore have unrestricted dual multipliers.

How a Solver Fits the Modeling Workflow

Interpret a candidate only after checking what the solver established:

1. **Read the status.** An optimal status reports optimality within numerical tolerances. A time or iteration limit does not; first confirm a feasible solution is available.
2. **Verify feasibility.** Check the original constraints and bounds, including every node balance for a flow model.
3. **Recompute the objective.** Use the returned quantities and original coefficients; compare with the reported value.
4. **Check an optimality bound.** Verify dual feasibility before comparing primal and dual objective values.

Numerical checks: use stated tolerances appropriate to the units and scales. See the [JuMP solution guide](#).

Supporting Algorithm: Revised Simplex

How does a solver find a candidate? The [revised-simplex notebook](#) develops three ideas:

- A **feasible basis** selects constraint columns that determine a corner of the feasible region.
- **Reduced costs** account for objective changes when a nonbasic variable increases and basic variables adjust.
- The **ratio test** limits the step so that the updated solution remains feasible.

The algorithm explains solver mechanics; our modeling work checks and interprets the result.

Summary

We formulated allocation models and used duality to interpret and verify their solutions.

Key takeaways

- **Formulating allocation models.** We represented purchases and flows using continuous variables, linear objectives, and resource or balance constraints.
- **Duality and resource values.** We used dual bounds and shadow prices to connect allocation decisions with the marginal value of resources.
- **Evaluating results.** We checked feasibility and objective values, and used agreement with a feasible dual bound to establish optimality.

A feasible allocation that reaches the dual bound is optimal.